Fractions & Decimals: Proportional Towers & Two-Dot Notation

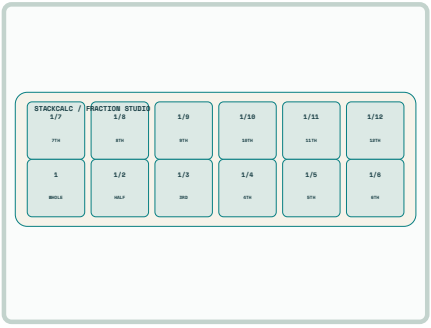
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.4.NF.A.2

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Fraction Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Compare a 3/4 tower (63mm) and a 5/8 tower (52.5mm). Observe 3/4 is taller.

Sketch Model on Paper:
Draw two equal bars: shade 3/4 and 5/8 to compare visually.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: FDISP Fraction Studio

1.	3/4	X: 3/4	Key 3/4 (.3.4) into
2.	ENTER	Y: 3/4 | X: 3/4	Stack lift
3.	5/8	Y: 3/4 | X: 5/8	Key 5/8 (.5.8) into
4.	-	X: 1/8	Subtract: 3/4 - 5/8

EXPECTED LCD X: 1/8 (0.1250)

1 **Compare 3/4 and 1/2. Subtract: 3/4 - 1/2.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Convert 7/10 to decimal on LCD: [7] [ENTER] [10] [÷].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Convert 50/100 to reduced fraction using [SIMP].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Add 5/10 and 35/100 as decimals.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 3

Fractions & Decimals: Proportional Towers & Two-Dot Notation • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Fraction Studio Towers (84.0mm Whole)

CAD Blueprint: fraction_tray

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/fraction_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Teal 3x1 Bin & Color-Coded Towers (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Compare 3/4 and 1/2. Subtract: 3/4 - 1/2.** Standard: CCSS.MATH.4.NF.A.2	1/4	[.] [3] [.] [4] [ENTER] [.] [1] [.] [2
2	**Convert 7/10 to decimal on LCD: [7] [ENTER]** Standard: CCSS.MATH.4.NF.A.2	0.70	[7] [ENTER] [1][0] [÷]
3	**Convert 50/100 to reduced fraction using [SI** Standard: CCSS.MATH.4.NF.A.2	1/2	[.] [5][0] [.] [1][0][0] [SIMP]
4	**Add 5/10 and 35/100 as decimals.** Standard: CCSS.MATH.4.NF.A.2	0.85 (85/100)	[0][.][5] [ENTER] [0][.][3][5] [+]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Do not compare fractions by looking only at numerators or denominators in isolation.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.